Corentin Léger - Machine Learning Research Engineer
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Research Engineer with 3 years of experience in Reinforcement Learning, Large Language Models and Multi-Agent systems across academic and industry settings. Interested in RL, LLMs and open-ended learning.

Skills

- **Technical**: Python, PyTorch, Ray (RLlib, Tune), Scikit-Learn, JAX, Gymnasium, Optuna, Hydra, Pydantic, CI/CD
- **Research**: Machine Learning, Reinforcement Learning, Large Language Models

Experience

- **InstaDeep** — Paris, France
  Research Engineer — *Oct 2025 - Present*
  - Reinforcement Learning research for real-world train dispatching optimization with Deutsche Bahn.
  - Designed and implemented a parallelized hyperparameter optimization pipeline (Ray, Optuna), achieving a 4× speedup over the previous workflow. Enabled reaching 100% scenarios solved in eval with a single agent approach.
  - Reduced mean train delays by 50% across single and multi-agent approaches without degrading other target metrics, by creating a multi-channel dense reward shaping as well as optimizing reward and PPO parameters.
  - Redesigned network architectures by implementing attention-based encoders over multi-train and route observations, as well as aggregating missing information in our PPO critic. This improved both sample efficiency and final policy performance.
  - Improved research infrastructure by finding and fixing a silent data-loss bug in our distributed rollout collection stack, and improving experiment pipeline robustness. Maintained code quality through code reviews and automated tests for CI/CD.

- **Huawei - Noah's Ark Lab** — Paris, France
  Research Engineer — *Jan 2025 - Oct 2025*
  - Automated Machine Learning and LLMs research at Paris Noah's Ark Lab, supervised by Balázs Kégl.
  - Developed a data preprocessing pipeline combining feature engineering, with LLMs code generation and heuristics, and feature selection methods to enhance the performance of an internal AutoML tool.
  - Created a post-training pipeline with TRL to fine-tune Llama3 8B on feature engineering code generation with GRPO.
  - Contributed to papers studying LLM hyperparameter generation skills (NeurIPS 2025 workshop) and bilevel optimization for Reinforcement Learning (preprint).

- **Inria - Flowers Lab** — Bordeaux, France
  Research Engineer — *Dec 2023 - Dec 2024*
  - LLMs and Multi-agent systems research at Flowers Lab, supervised by Clément Moulin Frier.
  - Contributed to two papers (ICLR 2025, preprint) studying text evolution in multi-turn LLM interactions. Built multi-agent simulation, NLP (SpaCy, NLTK) and data visualization tools to run experiments and evaluate metrics (LLM-Culture).
  - Co-developed a framework enabling evolutionary prompt optimization for multi-agent debate (NeurIPS 2025 workshop).
  - Co-developed Vivarium, an interactive multi-agent simulator built in Jax. Achieved real-time communication with Web or Jupyter notebook clients, and used the simulator in a Master's course at UPF Barcelona.

  Research Scientist Intern — *May 2023 - Nov 2023*
  - Meta-Reinforcement Learning research at Flowers Lab, supervised by Clément Moulin Frier
  - Published ER-MRL (EvoAPPS 2024), a method optimizing RNNs with Evolutionary Strategies, in order to improve Deep RL agents' abilities. It enabled to solve partially observable tasks, and adapt faster to unseen environments (Sb3, Gym, Optuna).
  - Co-authored a paper studying the combination of Tabular Reinforcement Learning and Symbolic data (HAL Inria).

- **Connectiv-IT** — Bordeaux, France
  Data Scientist Intern — *May 2022 - Aug 2022*
  - Preprocessed helicopter maintenance data, performing cleaning, outlier detection and imputation (Pandas, Scikit-Learn). Used statistical analysis (SciPy) and clustering to identify key trends in maintenance data.

Education

- **Ecole Nationale Supérieure de Cognitique** — Bordeaux, France
  Master of Science in Computer and Cognitive Sciences — *Sept. 2020 – Sept. 2023*
  Activities: Bronze medal at French University Volley-Ball Championship 2023

- **Cycle Préparatoire de Bordeaux** — Bordeaux, France
  Bachelor of Science in Mathematics and Physics, Sport-Study contract with ASI Volley-ball — *Sept. 2018 – Jun. 2020*